

JDScience 11+ Mathematics Practice Paper 1

Mathematics · Practice Paper 1

Name _____

Date _____

Score _____

Recommended time: 50 minutes

Questions: 40

Calculator: Not required unless a question says otherwise

Instructions

1. Work carefully and show clear thinking. For multiple-choice questions, choose the best answer.
2. If you cannot do a question, move on and return to it later if you have time.
3. Check your answers if you finish early.
4. This Mathematics paper is an original JDScience practice resource for 11+ preparation.
5. Each question has five options labelled A to E. Choose one answer only.

This is an original JDScience practice paper written for school entrance exam preparation. It is not an official GL Assessment, CEM, CSSE, school or publisher paper.

Mathematics

1. What is $3\,487 + 2\,596$?
 - A. 5 973
 - B. 6 083
 - C. 6 173
 - D. 5 983
 - E. 6 093
2. Calculate $8\,052 - 3\,679$.
 - A. 4 273
 - B. 4 373
 - C. 4 483
 - D. 5 373
 - E. 4 383
3. What is 47×36 ?
 - A. 1 592
 - B. 1 682
 - C. 1 692
 - D. 1 702
 - E. 1 582
4. Divide 2 448 by 12.
 - A. 194
 - B. 204
 - C. 214
 - D. 224
 - E. 234
5. Which number is a multiple of both 8 and 12?
 - A. 16
 - B. 24
 - C. 36
 - D. 40
 - E. 44

6. What is the highest common factor (HCF) of 36 and 54?

- A. 6
- B. 9
- C. 12
- D. 18
- E. 27

7. Round 47 385 to the nearest thousand.

- A. 47 000
- B. 47 400
- C. 48 000
- D. 47 390
- E. 50 000

8. What is $\frac{3}{5}$ of 85?

- A. 48
- B. 51
- C. 55
- D. 60
- E. 68

Mathematics

9. Calculate $2\frac{1}{4} + 1\frac{3}{8}$. Give your answer as a mixed number.

- A. $3\frac{1}{2}$
- B. $3\frac{5}{8}$
- C. $3\frac{7}{8}$
- D. $4\frac{1}{8}$
- E. $3\frac{3}{4}$

10. Simplify the fraction $\frac{42}{56}$.

- A. $\frac{3}{4}$
- B. $\frac{5}{7}$
- C. $\frac{6}{8}$
- D. $\frac{7}{8}$
- E. $\frac{2}{3}$

11. What is 0.75 as a fraction in its simplest form?

- A. $\frac{75}{100}$
- B. $\frac{3}{4}$
- C. $\frac{7}{10}$
- D. $\frac{15}{20}$
- E. $\frac{6}{8}$

12. Calculate 4.6×0.3 .

- A. 0.138
- B. 1.38
- C. 13.8
- D. 0.18
- E. 1.18

13. What is $7.2 \div 0.8$?

- A. 0.9
- B. 8
- C. 9
- D. 90
- E. 0.09

- 14.** Find 35% of £60.
- A. £18
 - B. £21
 - C. £24
 - D. £15
 - E. £25
- 15.** A jumper costs £40. In a sale it is reduced by 15%. What is the sale price?
- A. £32
 - B. £34
 - C. £35
 - D. £36
 - E. £28
- 16.** A shopkeeper buys a lamp for £25 and sells it for £40. What is the percentage profit?
- A. 15%
 - B. 40%
 - C. 50%
 - D. 60%
 - E. 75%

Mathematics

- 17.** Share £72 in the ratio 2 : 7. How much does the larger share receive?
- A. £16
 - B. £48
 - C. £56
 - D. £54
 - E. £63
- 18.** A recipe for 4 people needs 300 g of flour. How much flour is needed for 10 people?
- A. 600 g
 - B. 750 g
 - C. 700 g
 - D. 800 g
 - E. 650 g
- 19.** A map uses a scale of 1 : 50 000. What real distance does 4 cm on the map represent?
- A. 0.2 km
 - B. 2 km
 - C. 20 km
 - D. 200 km
 - E. 4 km
- 20.** A rectangle measures 12 cm by 7 cm. What is its perimeter?
- A. 19 cm
 - B. 38 cm
 - C. 42 cm
 - D. 84 cm
 - E. 26 cm
- 21.** A square has an area of 81 cm². What is the length of one side?
- A. 8 cm
 - B. 9 cm
 - C. 10 cm
 - D. 18 cm
 - E. 40.5 cm

- 22.** A triangular prism has a triangular face with base 6 cm and height 4 cm. The length of the prism is 10 cm. What is the volume?
- A. 120 cm^3
 - B. 240 cm^3
 - C. 60 cm^3
 - D. 100 cm^3
 - E. 48 cm^3
- 23.** An angle in a triangle is 47° and another is 68° . What is the third angle?
- A. 55°
 - B. 65°
 - C. 75°
 - D. 85°
 - E. 45°
- 24.** What is the size of each interior angle of a regular hexagon?
- A. 108°
 - B. 120°
 - C. 135°
 - D. 140°
 - E. 150°

Mathematics

- 25.** A train leaves at 09:45 and arrives at 12:20. How long is the journey?
- A. 2 hours 25 minutes
 - B. 2 hours 35 minutes
 - C. 2 hours 45 minutes
 - D. 3 hours 15 minutes
 - E. 2 hours 15 minutes
- 26.** A film starts at 14:30 and lasts 1 hour 50 minutes. At what time does it finish?
- A. 15:20
 - B. 16:00
 - C. 16:20
 - D. 16:10
 - E. 15:50
- 27.** Convert 3.65 kg into grams.
- A. 365 g
 - B. 3 650 g
 - C. 36 500 g
 - D. 365.0 g
 - E. 3.65 g
- 28.** A bottle holds 1.2 litres. How many millilitres is this?
- A. 12 ml
 - B. 120 ml
 - C. 1 200 ml
 - D. 12 000 ml
 - E. 0.12 ml
- 29.** Maya buys 3 notebooks at £1.85 each and pays with a £10 note. How much change does she receive?
- A. £4.45
 - B. £4.55
 - C. £5.45
 - D. £4.35
 - E. £5.55

- 30.** A bag of apples costs £2.40 for 6 apples. What is the cost of 15 apples at the same rate?
- A. £5.40
 - B. £6.00
 - C. £6.40
 - D. £7.20
 - E. £4.80
- 31.** The mean of five numbers is 12. Four of the numbers are 8, 10, 14 and 15. What is the fifth number?
- A. 11
 - B. 12
 - C. 13
 - D. 14
 - E. 15
- 32.** In a class of 30 pupils, 12 walk to school, 10 travel by bus and the rest cycle. What fraction of the class cycles?
- A. $\frac{1}{3}$
 - B. $\frac{4}{15}$
 - C. $\frac{1}{5}$
 - D. $\frac{2}{5}$
 - E. $\frac{3}{10}$

Mathematics

- 33.** A bar chart shows the number of books read by five children: Amira 6, Ben 4, Cora 9, Dylan 3, Ella 8. What is the median number of books?
- A. 4
 - B. 5
 - C. 6
 - D. 7
 - E. 8
- 34.** A spinner has 8 equal sections: 3 red, 2 blue and 3 green. What is the probability of landing on blue?
- A. $\frac{1}{8}$
 - B. $\frac{1}{4}$
 - C. $\frac{2}{5}$
 - D. $\frac{3}{8}$
 - E. $\frac{1}{2}$
- 35.** A rectangular garden is 18 m long and 12 m wide. A path of width 1 m runs around the inside of the garden. What is the area of the path?
- A. 54 m^2
 - B. 56 m^2
 - C. 58 m^2
 - D. 60 m^2
 - E. 62 m^2
- 36.** Tom has £48. He spends $\frac{1}{3}$ of his money on a book and then $\frac{1}{4}$ of what remains on a snack. How much money does he have left?
- A. £20
 - B. £24
 - C. £28
 - D. £18
 - E. £22
- 37.** A car travels 180 km in 3 hours at a constant speed. How far would it travel in 5 hours at the same speed?
- A. 240 km
 - B. 280 km
 - C. 300 km
 - D. 320 km
 - E. 360 km

- 38.** What is the next number in the sequence: 3, 7, 15, 31, ...?
- A. 47
 - B. 55
 - C. 63
 - D. 62
 - E. 59
- 39.** A cube has edges of 5 cm. What is its surface area?
- A. 25 cm^2
 - B. 100 cm^2
 - C. 125 cm^2
 - D. 150 cm^2
 - E. 30 cm^2
- 40.** Priya saves £3.50 each week. How many weeks will it take her to save at least £50?
- A. 12
 - B. 13
 - C. 14
 - D. 15
 - E. 16